# HorizonLink-SMD

0.1d

## Table of contents

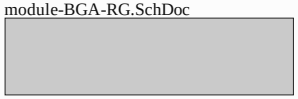

Sheet 1: Project overview (this page)

Sheet 2: module-BGA-RG

Device Sheets:

- Sheet 3: stm32wl55jci7-details
- Sheet 4: STM32RF_power_SMPS
- Sheet 5: CLS_1W_3.3V_Lineup


U_module-BGA-RG
module-BGA-RG.SchDoc



## Legend

General comment such as function title, configuration, ...

Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

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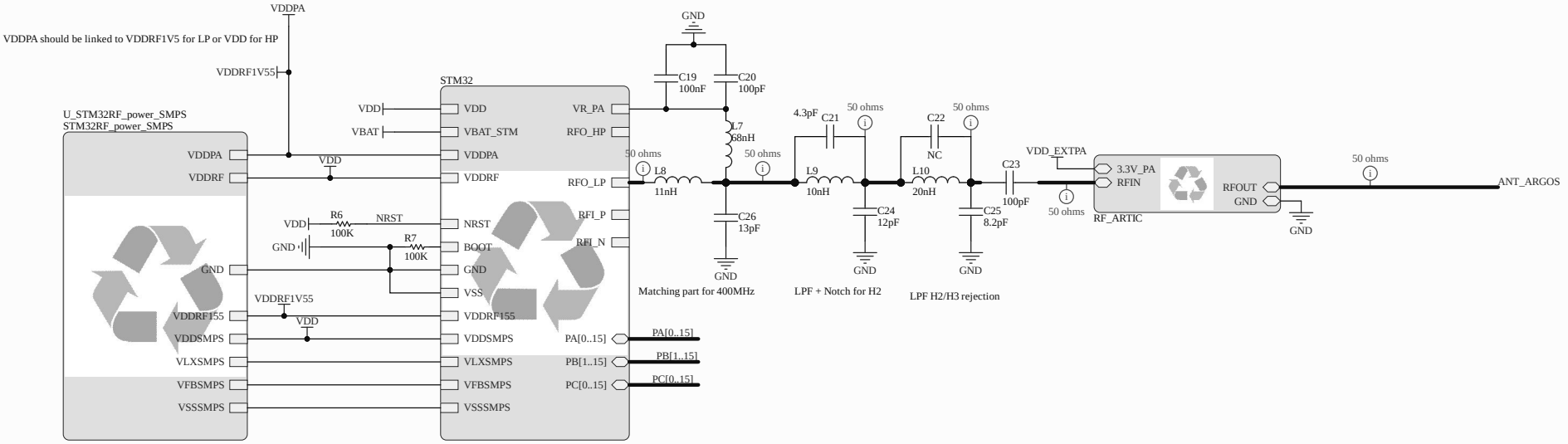
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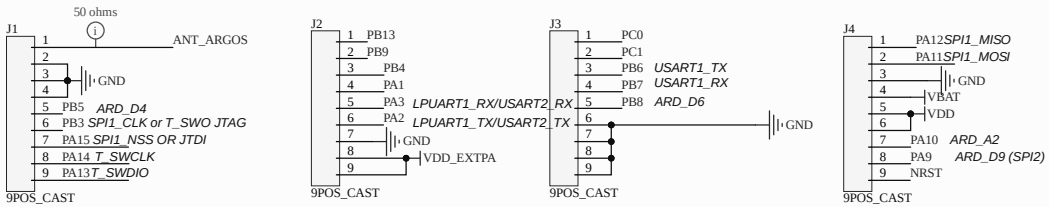
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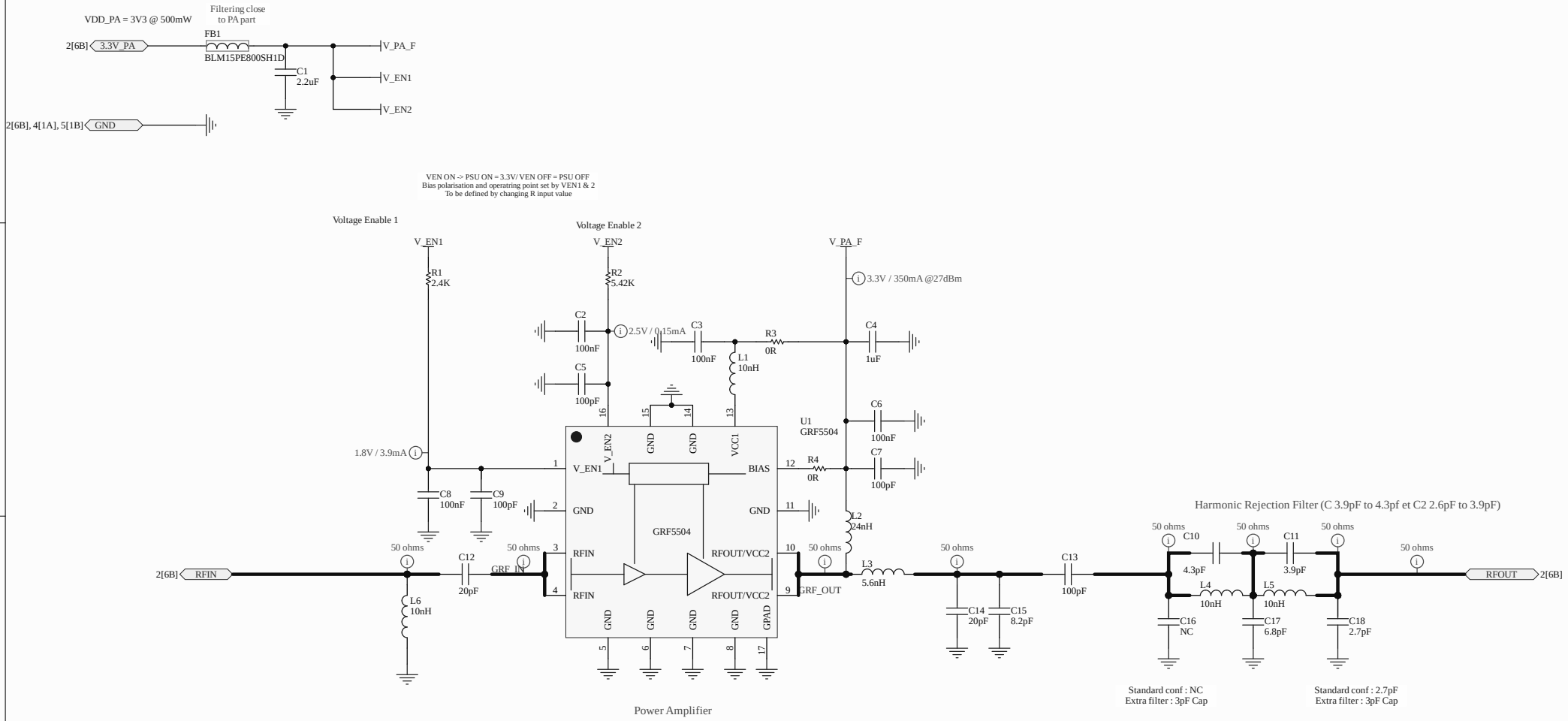
| | | | |
|---|---|---|---|
| Title ***HorizonLink-SMD*** | | | *Arribada Initiative*<br>*https://github.com/arribada*<br>*contact@arribada.org*<br>*arribada.org*<br>*UK - Portsmouth* |
| Size: A4 | Reference: 0.1d | Revision: D 0.1 | |
| Date: 19/03/2024 Time: 07:56:04 | | Sheet 1 of 5 | |
| File: D:\Altium\Projects\stm32wl-dev\Projects\BGA_RF-ArticGR\Project_Overview.SchDoc | | | |

VDD = 1.8 to 3.6V : External supply for system
VDDSMPS = 1.8V to 3.6V : External supply for SMPS Step down converter
VFBSMPS = 1.45V to 1.62V (1.55v) : External supply main system regulator
VDDA = 0 to 3.6V : 1.71V DAC min, 1.8V buffer, ADC 1.62V  VREFBUF = 2.24V
VDDRF = 1.8 to 3.6V :external power supply for radio (VDD)
VDDRF1V = 1.45V to 1.62V for radio
VBAT = 1.55V to 3.6V : RTC Tamp, external clock...
VREF-VREF+: input ref voltage for ADC



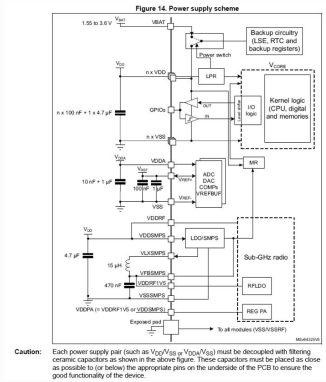

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| File: D:\Altium\Projects\stm32wl-dev\Projects\BGA_RF-ArticGR\module-BGA-RG.SchDoc | | | |




Notes: Murrata
Inductor from LQG ceramic 0402
Inductor from LQW wirewound 0603
Capacitor GRM/GJM 0402

50 ohm Line

| | | | |
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| File: D:\Altium\Projects\stm32wl-dev\DevicesSheets\RF\CLS_1W_3.3V_Lineup.SchDoc | | | |




**Caution:** Each power supply pair (such as $V_{DD}/V_{SS}$ or $V_{DDA}/V_{SSA}$) must be decoupled with filtering ceramic capacitors as shown in the above figure. These capacitors must be placed as close as possible to (or below) the appropriate pins on the underside of the PCB to ensure the good functionality of the device.

GND

PA[0..15]

PB[1..15]

PC[0..15]

RF power part output (not managed here)

U2B

VDDSMPS — A2 VDDSMPS

VLXSMPS — B1 VLXSMPS

VFBSMPS — B2 VFBSMPS

VSSSMPS — A1 VSSSMPS

STM32WL55JCU6

VDDRF155 — F7 VDDRF1V55

VDDPA — H8 VDDPA

VDDRF — E8 VDDRF

VSS — G6 G7 G8 H6 VSSRF

U2A

VR_PA — H9 VR_PA

RFO_HP — J9 RFO_HP

RFO_LP — J8 RFO_LP

RFI_P — J6 RFI_P

RFI_N — H7 RFI_N

B1 1000u@100MHz

VDD_TCXO — F6

PB0-VDD_TCXO

OSC_IN — F8

OSC_OUT — G9

STM32WL55JCU6

X1

VCC 4 — OUT 3 TCXO_OUT

N.C. 1 — GND 2

R8 220R

C44 10pF

R9 10K

C45 1uF

C46 100nF

GND

TG2520SMN_32.0000M-MCGNNM0

TCXO 32 MHz 1.8-3.6V 0.5ppm

U2C

C47 5.1pF — XTAL_IN — B6 PC14-OSC32_IN

C48 5.1pF — XTAL_OUT — C5 PC15-OSC32_OUT

Y1 NX2012SA-32.768K-STD-MUB-1

PH3-BOOT0 — J5 BOOT

NRST — F5 NRST

STM32WL55JCU6

C49 100nF

U3 ESDALC6V1-1U2

GND

It is highly recommended to apply conformal coatings to the PCB area shown in Figure 13, especially for the LSE quartz, CL1, CL2, and paths to the OSC_IN and OSC_OUT pads as a protection against moisture, dust, humidity, and temperature extremes that may lead to startup problems.

VDD

C35 2.2uF — C36 100nF — C37 100nF — C38 100nF — C39 100nF

VDD Decoupling

close to pins A7, E2, E9, H5

GND

U2D

A7 VDD — E2 VDD — E9 VDD — H5 VDD

VSS B7 — VSS D9 — VSS E3 — VSS E5

GND

VBAT_STM

A8 VBAT

A5 VDDA

B5 VREF+

VBAT decoupling

close to pin A8

C40 1uF — C41 100nF

GND

STM32WL55JCU6

FB2 600R@100MHz

C42 2.2uF — C43 100nF

VDDA decoupling

close to pin A5

GND

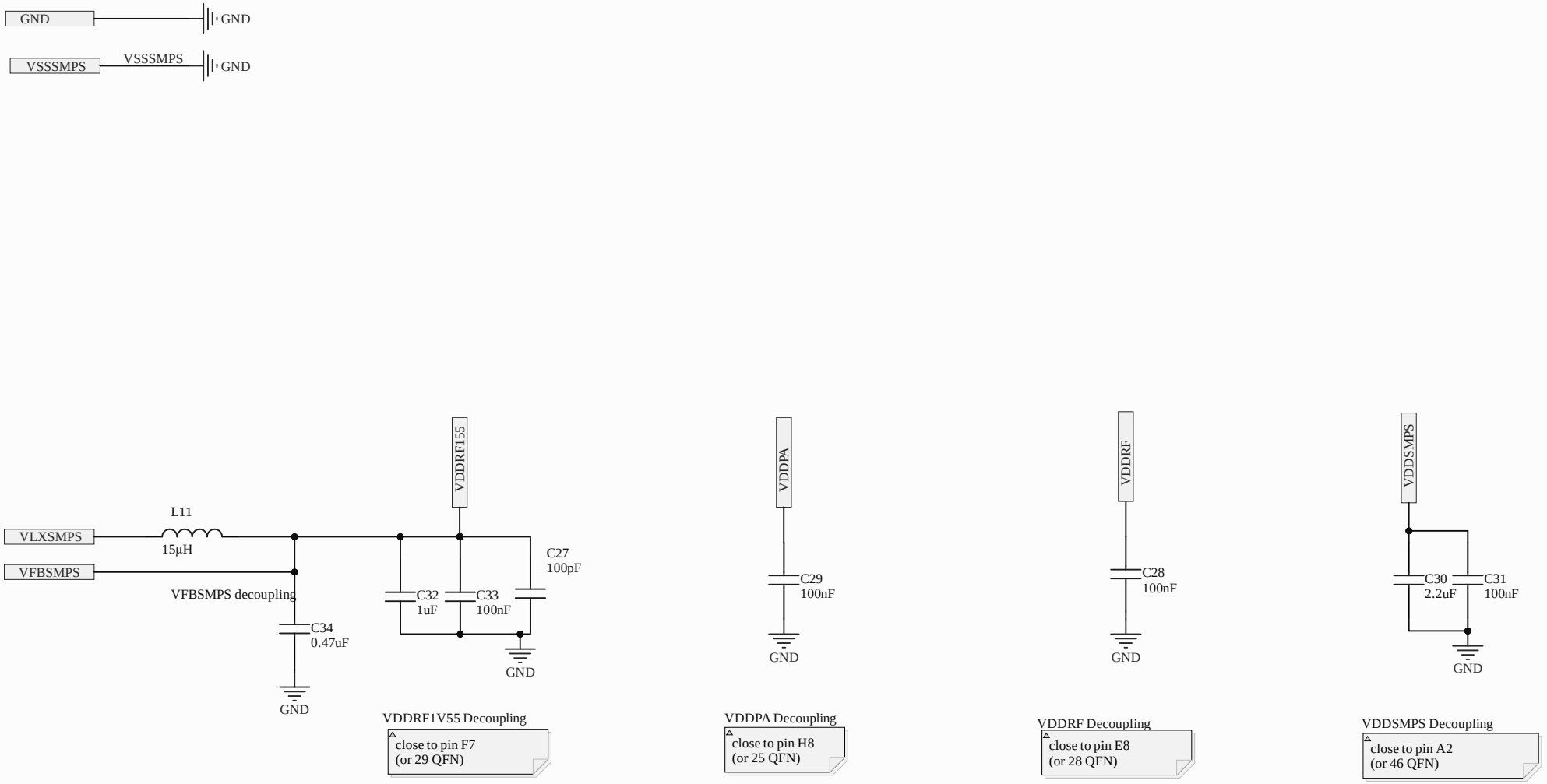
U2E

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| *B1* | PA0 | D6 | PA0 | PB1 | E7 | PB1 | *ARD_A0* |
| *B2* | PA1 | G3 | PA1 | PB2 | D8 | PB2 | *ARD_A1* |
| *LPUART1_TX/USART2_TX* | PA2 | H2 | PA2 | PB3 | C1 | PB3 | *ARD_D3/T_SWO* |
| *LPUART1_RX/USART2_RX* | PA3 | H1 | PA3 | PB4 | C2 | PB4 | *ARD_A3* |
| *ARD_D10* | PA4 | J1 | PA4 | PB5 | D2 | PB5 | *ARD_A4* |
| *ARD_D13* | PA5 | J2 | PA5 | PB6 | E1 | PB6 | *USART1_TX* |
| *ARD_D12* | PA6 | F4 | PA6 | PB7 | C3 | PB7 | *USART1_RX* |
| *ARD_D11* | PA7 | H3 | PA7 | PB8 | D3 | PB8 | ***ARD_D6*** |
| *MCO* | PA8 | J3 | PA8 | PB9 | C4 | PB9 | |
| *ARD_D9* | PA9 | E5 | PA9 | PB10 | H4 | PB10 | *ARD_D6* |
| *ARD_A2* | PA10 | C8 | PA10 | PB11 | G4 | PB11 | |
| *ARD_D14* | PA11 | B9 | PA11 | PB12 | E6 | PB12 | *ARD_D2* |
| *ARD_D15* | PA12 | A9 | PA12 | PB13 | D7 | PB13 | *ARD_A5* |
| *T_SWDIO* | PA13 | B8 | PA13 | PB14 | C6 | PB14 | *ARD_A4* |
| *T_SWCLK* | PA14 | A4 | PA14 | PB15 | B4 | PB15 | |
| *T_JTDI* | PA15 | B3 | PA15 | | | | |
| | PC0 | F2 | PC0 | | | | |
| *ARD_D7* | PC1 | F1 | PC1 | | | | |
| *ARD_D8* | PC2 | D4 | PC2 | | | | |
| | PC3 | D5 | PC3 | | | | |
| | PC4 | F3 | PC4 | | | | |
| | PC5 | E4 | PC5 | | | | |
| *B3* | PC6 | G2 | PC6 | | | | |
| | PC13 | C7 | PC13 | | | | |

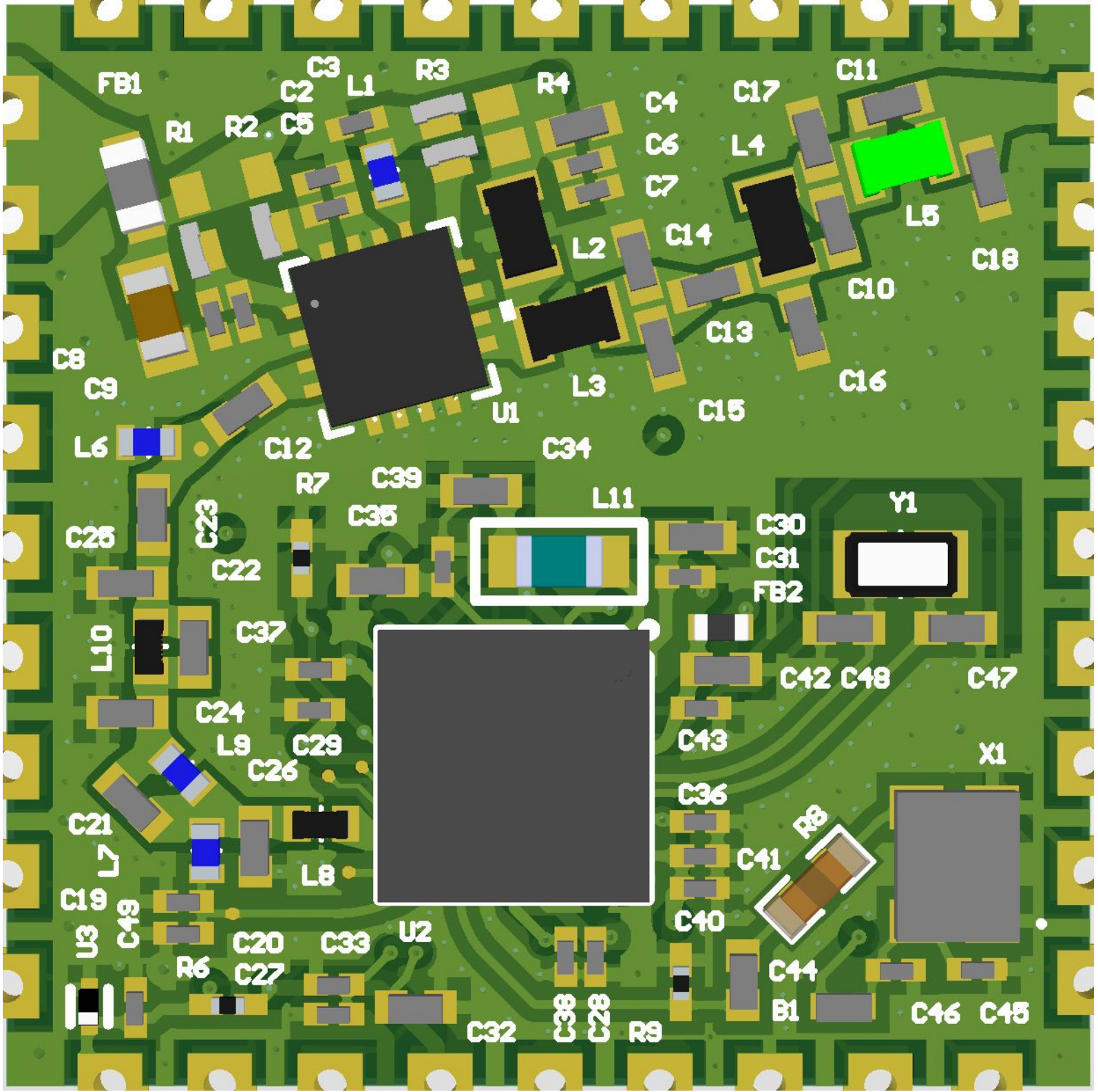
STM32WL55JCU6

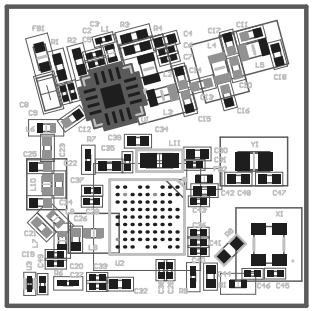
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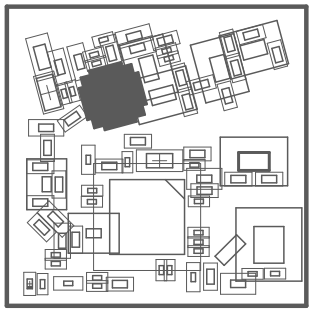



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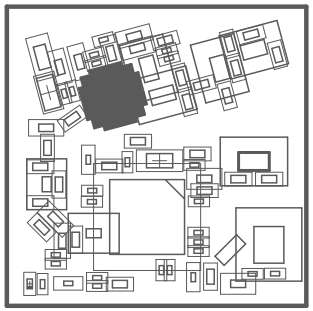

FB1
C3
C2
L1
R3
R4
C11
C17
C4
C6
C7
L4
R1
R2
C5
L5
C14
L2
C18
C10
C13
C8
C9
L3
C16
U1
C15
L6
C12
C34
R7
C39
L11
Y1
C23
C35
C30
C31
FB2
C25
C22
C37
L10
C42
C48
C47
C24
L9
C29
C26
C43
X1
C36
C21
L7
L8
R8
C41
C19
C49
U3
C40
C20
C33
U2
C44
R6
C27
B1
C46
C45
C32
C38
C28
R9



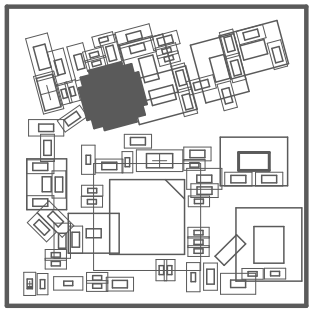
| Layer | Name | Material | Thickness | Constant | Board Layer Stack |
|---|---|---|---|---|---|
| | Top Overlay | | | | |
| | Top Solder | Solder Resist | 0,020mm | 3.5 | |
| 1 | Top Layer | | 0,035mm | | |
| | Dielectric 1 | 1080HR RC68 | 0,076mm | 4.18 | |
| 2 | Signal Layer 1 | | 0,035mm | | |
| | Dielectric 2 | 7 x 7628 | 1,268mm | 4.74 | |
| 3 | Signal Layer 2 | | 0,035mm | | |
| | Dielectric 3 | 1080HR RC68 | 0,076mm | 4.18 | |
| 4 | Bottom Layer | | 0,035mm | | |
| | Bottom Solder | Solder Resist | 0,020mm | 3.5 | |
| | Bottom Overlay | | | | |
| Total board thickness: | | | 1,600mm | | |



| Layer | Name | Material | Thickness | Constant | Board Layer Stack |
|---|---|---|---|---|---|
| | Top Overlay | | | | |
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| 3 | Signal Layer 2 | | 0,035mm | | |
| | Dielectric 3 | 1080HR RC68 | 0,076mm | 4.18 | |
| 4 | Bottom Layer | | 0,035mm | | |
| | Bottom Solder | Solder Resist | 0,020mm | 3.5 | |
| | Bottom Overlay | | | | |
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| Layer | Name | Material | Thickness | Constant | Board Layer Stack |
|---|---|---|---|---|---|
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| | Bottom Solder | Solder Resist | 0,020mm | 3.5 | |
| | Bottom Overlay | | | | |
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| Layer | Name | Material | Thickness | Constant | Board Layer Stack |
|---|---|---|---|---|---|
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| | Dielectric 2 | 7 x 7628 | 1,268mm | 4.74 | |
| 3 | Signal Layer 2 | | 0,035mm | | |
| | Dielectric 3 | 1080HR RC68 | 0,076mm | 4.18 | |
| 4 | Bottom Layer | | 0,035mm | | |
| | Bottom Solder | Solder Resist | 0,020mm | 3.5 | |
| | Bottom Overlay | | | | |
| Total board thickness: | | | 1,600mm | | |

| Line # | Name | Description | Designator | Quantity | Manufacturer 1 | Manufacturer Part Number 1 | Manufacturer Lifecycle 1 | Supplier 1 | Supplier Part Number 1 | Supplier Unit Price 1 | Supplier Subtotal 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1000u@100MHz | Ferrite Bead - BLM15_Z1D Series EMI Suppression Filter 1000+/-25% at 100MHz 250mA @85 | B1 | 1 | Murata | BLM15HG102SZ1D | Volume Production | | | | |
| | 2.2uF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.2 uF 10 VDC 10% 0603 X7R | C1 | 1 | Murata | GRM188R71A225KE15D | Volume Production | Mouser | 81-GRM188R71A225KE15 | | |
| | 100nF | Multilayer Ceramic Capacitors MLCC - SMD/SMT .1UF 16V 10% 0201 | C2, C3, C6, C8, C19, C31, C33, C36, C37, C38, C39, C41, C43, C46, C49 | 15 | Murata | GRM033Z71C104KE14D | Volume Production | Mouser | 81-GRM033Z71C104KE4D | | |
| | 1uF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 1.0UF 10V 10% 0402 | C4, C32 | 2 | Murata | GRM155Z71A105KE01J | Volume Production | Mouser | 81-GRM155Z71A105KE1J | | |
| | 100pF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 100PF 25V 5% 0201 | C5, C7, C9, C20 | 4 | Murata | GRM0335C1E101JA01D | Volume Production | Mouser | 81-GRM0335C1E101JA1D | | |
| | GRM1555C1H4R3BA01D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 4.3 pF 50 VDC 0.1 pF 0402 C0G (NP0) | C10, C21 | 2 | Murata | GRM1555C1H4R3BA01D | | Digi-Key | 490-6243-1-ND | | |
| | GRM1555C1E3R9BA01D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 3.9 pF 25 VDC 0.1 pF 0402 C0G (NP0) | C11 | 1 | Murata | GRM1555C1E3R9BA01D | | Mouser | 81-GRM1555C1E3R9BA1D | | |
| | 20pF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 20 pF 50 VDC 2% 0402 C0G (NP0) | C12, C14 | 2 | Murata | GRM1555C1H200GA01D | Not Recommended for New Design | Mouser | 81-GRM1555C1H200GA1D | | |
| | GCM1555C1H101JA16D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 100 pF 50 VDC 5% 0402 C0G (NP0) AEC-Q200 | C13, C23 | 2 | Murata | GCM1555C1H101JA16D | | Mouser | 81-GCM1555C1H101JA6D | | |
| | GCM1555C1H8R2CA16J | Multilayer Ceramic Capacitors MLCC - SMD/SMT 8.2 pF 50 VDC 0.25 pF 0402 C0G (NP0) AEC-Q200 | C15, C25 | 2 | Murata | GCM1555C1H8R2CA16J | Volume Production | Mouser | 81-GCM1555C1H8R2CA6J | | |
| | CAP-NC-0402 | CMS0402 Non Connected | C16, C22 | 2 | | | | | | | |
| | GRM1555C1E6R8BA01D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 6.8 pF 25 VDC 0.1 pF 0402 C0G (NP0) | C17 | 1 | Murata | GRM1555C1E6R8BA01D | | Mouser | 81-GRM1555C1E6R8BA1D | | |
| | GRM1555C2A2R7CA01D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.7 pF 100 VDC 0.25 pF 0402 C0G (NP0) | C18 | 1 | Murata | GRM1555C2A2R7CA01D | | Mouser | 81-GRM1555C2A2R7CA1D | | |
| | GCM1555C1H120GA16D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 12 pF 50 VDC 2% 0402 C0G (NP0) AEC-Q200 | C24 | 1 | Murata | GCM1555C1H120GA16D | Volume Production | Mouser | 81-GCM1555C1H120GA6D | | |
| | GRM1555C1H130GA01D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 13 pF 50 VDC 2% 0402 C0G (NP0) | C26 | 1 | Murata | GRM1555C1H130GA01D | Not Recommended for New Design | Mouser | 81-GRM1555C1H130GA1D | | |
| | 100pF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 100 pF 25 VDC 1% 0201 C0G (NP0) | C27 | 1 | Murata | GRM0335C1E101FA01J | Volume Production | Digi-Key | GRM0335C1E101FA01J-ND | | |
| | GRM033R61C104KE14D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 0201 0.1uF 16volts X5R 10% | C28, C29 | 2 | Murata | GRM033R61C104KE14D | | Mouser | 81-GRM033R61C104KE4D | | |
| | GRM155Z71A225KE44D | Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.2 uF 10 VDC 10% 0402 X7R | C30, C35, C42 | 3 | Murata | GRM155Z71A225KE44D | Volume Production | Mouser | 81-GRM155Z71A225KE4D | | |
| | 0.47uF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.47uF 10V X5R 10% SMD 0402 85°C Paper T/R | C34 | 1 | Taiyo Yuden | LMK105ABJ474KV8F | | Mouser | 963-LMK105ABJ474KV8F | | |
| | 1uF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 1uF+/-10% 10V X5R 06 0201 | C40, C45 | 2 | Samsung | CL03A105KP3NSNC | Volume Production | | | | |
| | 10pF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 10 pF 50 VDC 1% 0402 C0G (NP0) | C44 | 1 | Murata | GJM1555C1H100FB01D | Volume Production | | | | |
| | 5.1pF | Multilayer Ceramic Capacitors MLCC - SMD/SMT 0402 MLCC NPO 5.1 pF, +/- 0.1pF 50 V T&RGP | C47, C48 | 2 | Walsin Technologies | 0402N5R1B500CT | | Mouser | 791-0402N5R1B500CT | | |
| | BLM15PE800SH1D | Ferrite Bead, 80 Ohm, 25%, 0402 Rohs Compliant: Yes | FB1 | 1 | Murata | BLM15PEB00SH1D | Volume Production | Digi-Key | 490-BLM15PEB00SH1DCT-ND | | |
| | 600R@100MHz | Ferrite Bead 600ohm 500mA 340mOhm 0402 | FB2 | 1 | Murata | BLM15BX601SZ1D | Volume Production | | | | |
| | 9POS_CAST | 9 POS Castellated connector with VIA 0.6 | J1, J2, J3, J4 | 4 | | | | | | | |
| | 10nH | Inductor RF Chip Multi-Layer 10nH 5% 100MHz 8Q-Factor Air 300mA 260mOhm DCR 0402 Paper T/R | L1, L6, L9 | 3 | Murata | LQG15HS10NJ02D | Volume Production | Arrow Electronics | LQG15HS10NJ02D | | |
| | 24nH | Inductor RF Chip Unshielded Wirewound 24nH 2% 100MHz 40Q-Factor Ferrite 1.3A 0603 Paper T/R | L2 | 1 | Murata | LQW18AN24NG80D | Volume Production | Mouser | 81-LQW18AN24NG80D | | |
| | LQW18AN5N6C80D | Inductor Wirewound , 5.6 nH, 40 Milliohm, ± 0.2nH, 6.65 GHz, 1.9 A, 0603 [1608 Metric] | L3 | 1 | Murata | LQW18AN5N6C80D | Volume Production | Mouser | 81-LQW18AN5N6C80D | | |
| | 10nH | Inductor Wirewound , 10 nH, 52 Milliohm, ± 2%, 4.75 GHz, 1.6 A, 0603 [1608 Metric] | L4, L5 | 2 | Murata | LQW18AN10NG80D | Volume Production | Digi-Key | 490-17899-1-ND | | |
| | LQG15HN68NJ02D | Inductor RF Chip Multi-Layer 68nH 5% 100MHz 8Q-Factor Air 180mA 1.25Ohm DCR 0402 Paper T/R | L7 | 1 | Murata | LQG15HN68NJ02D | Not Recommended for New Design | Digi-Key | 490-1095-1-ND | | |
| | LQG15WZ11NJ02D | Inductor RF Chip Multi-Layer 11nH 5% 0402 Paper T/R | L8 | 1 | Murata | LQG15WZ11NJ02D | Volume Production | Digi-Key | 490-15246-1-ND | | |
| | LQG15WH20NJ02D | Inductor RF Chip, Aec-Q200, 20Nh, 0402, 2.1Ghz Rohs Compliant: Yes | L10 | 1 | Murata | LQG15WH20NJ02D | Volume Production | Mouser | 81-LQG15WH20NJ02D | | |
| | MLZ1608M150WT000 | Inductor RF Shielded 15µH, ±5% | L11 | 1 | TDK | MLZ1608M150WT000 | Volume Production | Mouser | 810-MLZ1608M150WT000 | | |
| | 2.4K | Thick Film Resistors - SMD 2.4 kOhms 62.5 mW 0402 1% | R1 | 1 | Yageo | RC0402FR-132K4L | | Mouser | 603-RC0402FR-132K4L | | |
| | 5.42K | Thin Film Resistors - SMD 5.42kOhm,0402,1%,50p pm,63mW,50V | R2 | 1 | KOA Speer | RN73H1ETP5421F50 | | Mouser | 660-RN73H1ETP5421F50 | | |
| | 0R | Resistors Thick Film - SMD 0402 0.063 W 0 Ohm Jumper | R3, R4 | 2 | Vishay | CRCW040200000ZDED | | Digi-Key | 541-0.0JCT-ND | | |
| | 100K | Resistors Thick Film - SMD 1/20watt 100Kohms 1% 200ppm | R6, R7 | 2 | Vishay | CRCW0201100KFNED | Volume Production | | | | |
| | 220R | Resistors Thick Film - SMD HP02 (0402) 1/10W 1% 220R T/R 10000 | R8 | 1 | Royal Electronic Factory | HP02WAF2200TCE | | Mouser | 303-HP02WAF2200TCE | | |
| | 10K | Resistor Thick Film Anti-Sulfurated Chip 0201 10kOhm 1% Paper T/R | R9 | 1 | Yageo | AF0201FR-0710KL | | Mouser | 603-AF0201FR-0710KL | | |
| | GRF5504 | RF Amplifier High Efficiency, 3.5Watt PA/Saturated PA, OP1dB>34.0 dBm; 0.4 - 0.5 GHz | U1 | 1 | Guerrilla RF | GRF5504 | | Mouser | 459-GRF5504 | | |
| | STM32WL55JCI7_details | Multiprotocol Wireless 32-bit MCU with LPWAN Sub-GHz radio solution(BGA) | U2 | 1 | STMicroelectronics | STM32WL55JCI7 | | Mouser | 511-STM32WL55JCI7 | | |
| | ESD_6V | ESD protection 6V1 - Varistors MLV 0603 18V 3pF | U3 | 1 | STMicroelectronics | ESDALC6V1-1U2 | Volume Production | Mouser | 511-ESDALC6V1-1U2 | | |
| | TG2520SMN_32.0000M-MCGNNM0 | TCXO Oscillators TG2520SMN 32.0000M-MCGNNM0: TCXO 2.8V 3.3V 0.5PPM -40 85C NON O/E 1K TR | X1 | 1 | Epson | TG2520SMN32.0000M-MCGNNM0 | | Mouser | 732-2520SMN32MCGNNM0 | | |
| | NX2012SA-32.768K-STD-MUB-1 | Crystals 32.768KHZ 12.5PF SMD | Y1 | 1 | NDK | NX2012SA-32.768K-STD-MUB-1 | Volume Production | Mouser | 344-NX2012SA32768K1 | | |